

Fashion-MNIST Image Classification Using Fully Connected Neural Networks

Ankit Saini(2522Ma01)
Tiyas Chakrabortty(2522Ma02)
Tanu Goliyan(2621Ma05)
Murari Ray(2621Ma07)
Department of Mathematics
IIT Patna

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Abstract

This report presents the implementation and evaluation of Fully Connected Neural Networks (FCNNs) for five-class image classification using the Fashion-MNIST dataset. Five classes, namely T-shirt/top, Trouser, Pullover, Dress, and Coat, were selected from the original dataset. Three different network architectures containing three, four, and five hidden layers were investigated. Six optimization algorithms, namely Stochastic Gradient Descent (SGD), Batch Gradient Descent, Momentum, Nesterov Accelerated Gradient (NAG), RMSProp, and Adam, were compared. A total of 18 experiments were performed. The best-performing model was the four-hidden-layer FCNN trained using SGD, which achieved a validation accuracy of 91.41% and a final test accuracy of 91.70%.

1 Introduction

Image classification is an important problem in machine learning and deep learning in which an input image is assigned to one of several predefined classes. Neural networks are widely used for this task because they can learn complex nonlinear relationships between input features and output classes.

In this experiment, a Fully Connected Neural Network (FCNN) was implemented for the classification of clothing images from the Fashion-MNIST dataset. Unlike convolutional neural networks, a fully connected network treats the image pixels as a vector of input features. Each input neuron is connected to neurons in the following layer.

The original Fashion-MNIST dataset contains ten different clothing categories. For this experiment, only five classes were considered: T-shirt/top, Trouser, Pullover, Dress, and Coat. Three FCNN architectures with different numbers of hidden layers were trained using six different optimization techniques.

The purpose of the experiment was to study how the depth of the network and the choice of optimization algorithm affect training error, convergence, classification accuracy, and generalization performance.

2 Objectives

The main objectives of this experiment are:

1. To implement a Fully Connected Neural Network for Fashion-MNIST image classification.
2. To construct a balanced five-class dataset from Fashion-MNIST.
3. To compare FCNNs containing three, four, and five hidden layers.
4. To compare the performance of six optimization algorithms: SGD, Batch Gradient Descent, Momentum, NAG, RMSProp, and Adam.
5. To investigate the convergence behavior of the different optimizers.
6. To compare training accuracy and validation accuracy for all architecture-optimizer combinations.
7. To identify the best-performing architecture and optimizer using validation accuracy.
8. To evaluate the selected model on an unseen test dataset using test accuracy and confusion matrices.

3 Dataset and Preprocessing

3.1 Fashion-MNIST Dataset

The Fashion-MNIST dataset was obtained using the Fashion-MNIST implementation provided by the `torchvision` library.

The original dataset contains 60,000 training images and 10,000 test images. Each image is a grayscale image of size 28×28 pixels.

For this experiment, five classes were selected:

Table 1: Selected Fashion-MNIST classes

Label	Class
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat

Each of the five selected classes contains 7,000 images. Therefore, the total number of selected images is

$$5 \times 7000 = 35000.$$

Hence, the final dataset consists of 35,000 images distributed equally among the five classes.

3.2 Data Split

A stratified split was used to ensure that all classes remained equally represented in the training, validation, and test sets.

The resulting dataset split was:

Table 2: Dataset split

Dataset	Number of Samples	Percentage
Training	22,400	64%
Validation	5,600	16%
Test	7,000	20%
Total	35,000	100%

The class distribution in all three subsets was perfectly balanced.

Table 3: Class distribution in training, validation, and test sets

Class	Training	Validation	Test
T-shirt/top	4480	1120	1400
Trouser	4480	1120	1400
Pullover	4480	1120	1400
Dress	4480	1120	1400
Coat	4480	1120	1400

3.3 Input Representation

Each image has dimensions 28×28 . Therefore, each image contains

$$28 \times 28 = 784$$

pixel values.

The images were represented as vectors in $\mathbb{R}^{784}$.

The pixel values were scaled to the range $[0, 1]$. Thus, the resulting input matrix had the shape

$$X \in \mathbb{R}^{35000 \times 784}.$$

3.4 Sample Training Images

Figure below shows examples of the training images used in the experiment.

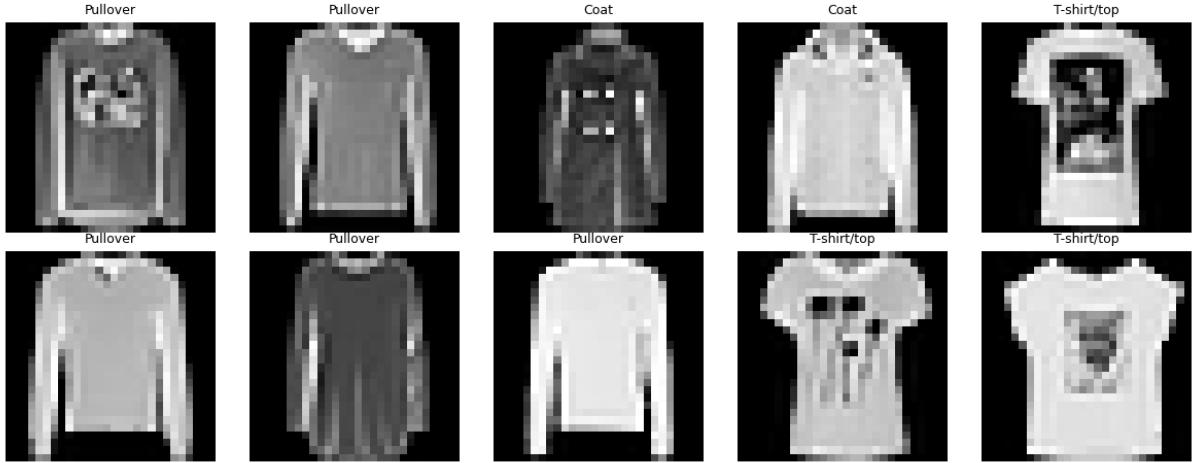


Figure 1: Sample Training Images

4 Methodology

4.1 Fully Connected Neural Network

A Fully Connected Neural Network was used for classification.

The input layer contains 784 neurons because each image contains 784 pixel values. The hidden layers use the ReLU activation function. The final layer contains five neurons corresponding to the five selected classes.

The three architectures used in the experiment were:

Table 4: FCNN architectures

Architecture	Hidden Layers	Parameters
3-hidden	128 $\rightarrow$ 64 $\rightarrow$ 32	110,981
4-hidden	128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16	111,429
5-hidden	128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8	111,525

For example, the best-performing architecture had the structure

$$784 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 5.$$

Each hidden layer applies a linear transformation followed by ReLU:

$$h = \text{ReLU}(Wx + b).$$

The final output layer produces scores for the five classes.

4.2 Loss Function

Cross-entropy loss was used as the training objective:

$$L = - \sum_i y_i \log(\hat{y}_i),$$

where y_i represents the true class and $\hat{y}_i$ represents the predicted class probability. In PyTorch, this was implemented using `CrossEntropyLoss`.

4.3 Optimization Algorithms

Six optimization algorithms were evaluated.

4.3.1 SGD

Standard Stochastic Gradient Descent was used with learning rate

$$\alpha = 0.001.$$

The batch size was 1.

4.3.2 Batch Gradient Descent

Batch Gradient Descent used the entire training set as one batch for each parameter update.

4.3.3 Momentum

Momentum was used with

$$\alpha = 0.001, \quad \mu = 0.9.$$

4.3.4 Nesterov Accelerated Gradient

NAG was used with

$$\alpha = 0.001, \quad \mu = 0.9.$$

4.3.5 RMSProp

RMSProp was configured using

$$\alpha = 0.001, \quad \beta = 0.99, \quad \epsilon = 10^{-8}.$$

4.3.6 Adam

Adam was configured using

$$\alpha = 0.001, \\ \beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon = 10^{-8}.$$

4.4 Initialization

A random seed of 42 was used. The initial weights were saved separately for each architecture and reused across the different optimizers so that the optimizer comparison started from identical initial weights for a given architecture.

4.5 Convergence Criterion

The convergence criterion was defined as

$$|E_t - E_{t-1}| < 0.001,$$

where E_t represents the average training error at epoch t .

Convergence checking was started only after five epochs. The maximum number of epochs was fixed at 100.

Therefore, the total number of experiments was

$$3 \times 6 = 18.$$

5 Experiment Results and Data Analysis

5.1 Training and Validation Results

The results obtained from all 18 experiments are presented in Table 5.

Table 5: Training and validation performance for all experiments

Architecture	Optimizer	Epochs	Train Acc.	Val. Acc.
4-hidden	SGD	30	95.19%	91.41%
5-hidden	SGD	29	94.99%	91.20%
4-hidden	NAG	22	93.47%	91.09%
3-hidden	SGD	40	95.30%	90.80%
4-hidden	Momentum	13	92.00%	90.57%
5-hidden	Momentum	30	92.61%	90.41%
3-hidden	Adam	35	92.81%	90.41%
3-hidden	Momentum	32	94.09%	90.36%
4-hidden	Adam	14	91.49%	89.80%
5-hidden	Adam	29	91.22%	89.59%
3-hidden	NAG	15	90.64%	88.96%
5-hidden	RMSProp	8	87.98%	87.57%
5-hidden	NAG	36	88.77%	86.77%
4-hidden	RMSProp	10	86.40%	85.96%
3-hidden	RMSProp	100	55.86%	55.41%
3-hidden	Batch GD	6	21.10%	20.66%
4-hidden	Batch GD	6	19.99%	20.00%
5-hidden	Batch GD	6	20.00%	20.00%

5.2 Convergence Table

The number of epochs required to satisfy the convergence criterion for different architecture-optimizer combinations is shown below.

Table 6: Epochs to convergence

Architecture	SGD	Batch GD	Momentum	NAG	RMSProp	Adam
3-hidden	40	6	32	15	100	35
4-hidden	30	6	13	22	10	14
5-hidden	29	6	30	36	8	29

5.3 Training Error Curves

The training error curves for all optimizers and architectures are shown in Figure ??.

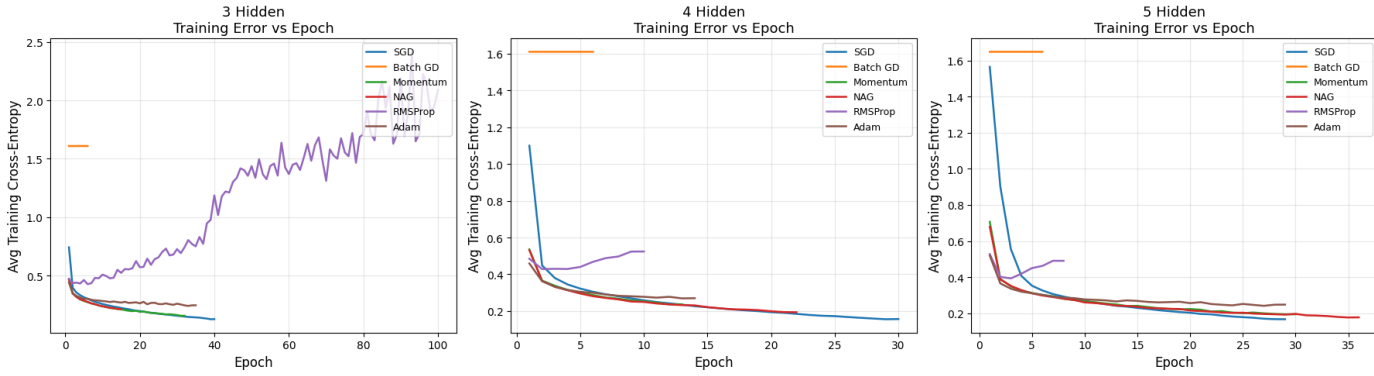


Figure 2: Training error plots

The error curves demonstrate substantially different optimization behaviors. SGD shows a consistent decrease in training error for the three architectures. Momentum and NAG also show effective reduction of the training error.

Adam reduces the training error relatively quickly in the early stages, although its final validation accuracy is lower than that obtained by SGD.

RMSProp displays unstable behavior for the 3-hidden-layer architecture. Its training error increases considerably during training and the model reaches only 55.86% training accuracy.

Batch Gradient Descent shows almost no useful learning under the selected settings.

5.4 Best Architecture and Optimizer

The best validation performance was obtained using the four-hidden-layer architecture with SGD.

The model achieved:

$$\text{Training Accuracy} = 95.19\%$$

$$\text{Validation Accuracy} = 91.41\%$$

and required 30 epochs to satisfy the convergence criterion. Therefore,

$$\boxed{\text{Best Model} = 4\text{-hidden-layer FCNN} + \text{SGD}}$$

5.5 Test Performance

After selecting the best model using validation accuracy, its performance was evaluated on the test set.

The resulting accuracies were:

Table 7: Final performance of the best model

Measure	Accuracy
Training Accuracy	95.19%
Validation Accuracy	91.41%
Test Accuracy	91.70%

The final test accuracy of

91.70%

indicates that the selected FCNN generalizes well to unseen samples.

5.6 Confusion Matrices

The confusion matrix for the training set is shown in Figure

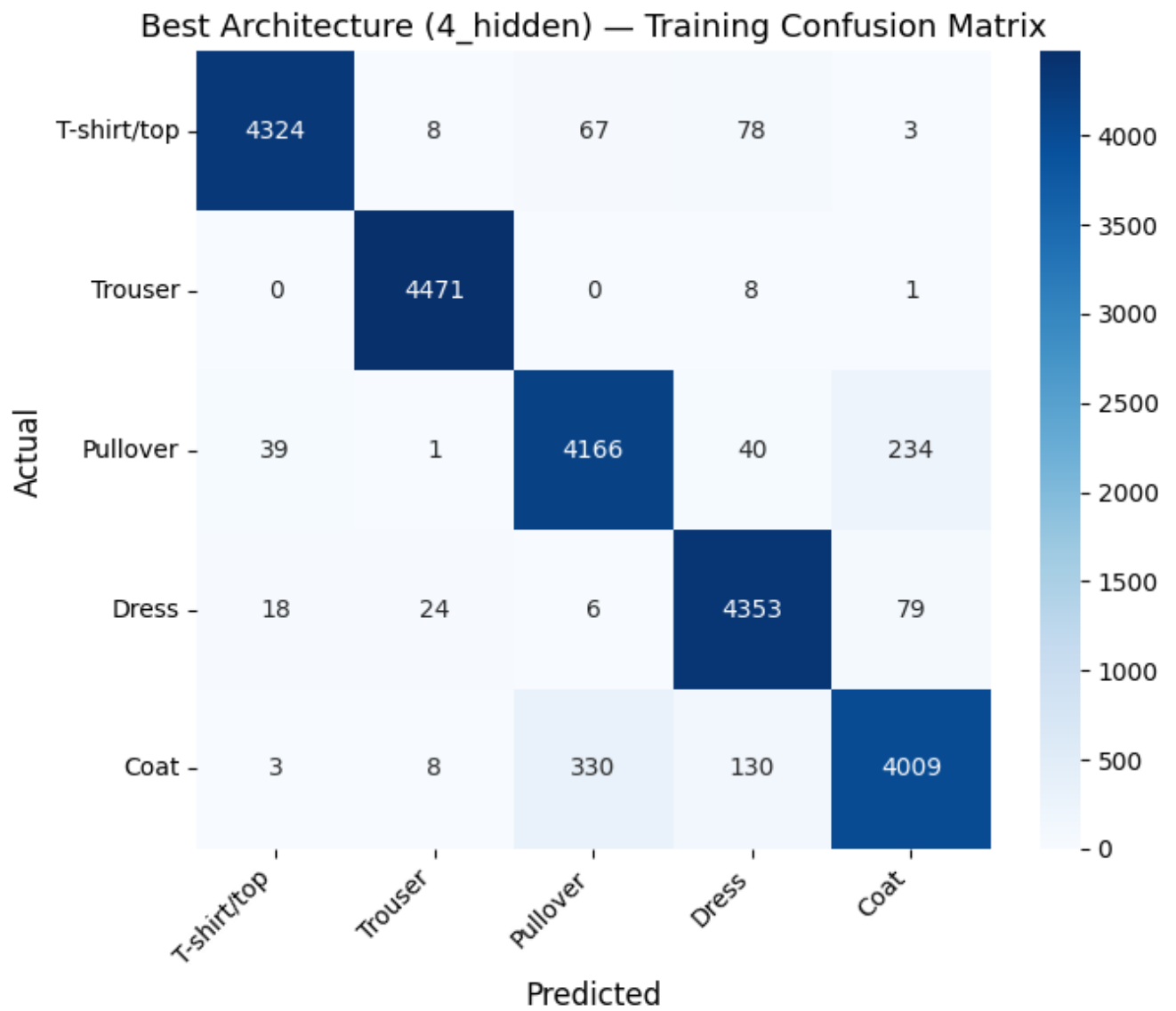


Figure 3: Training Confusing Matrix

The confusion matrix for the test set is shown in Figure.

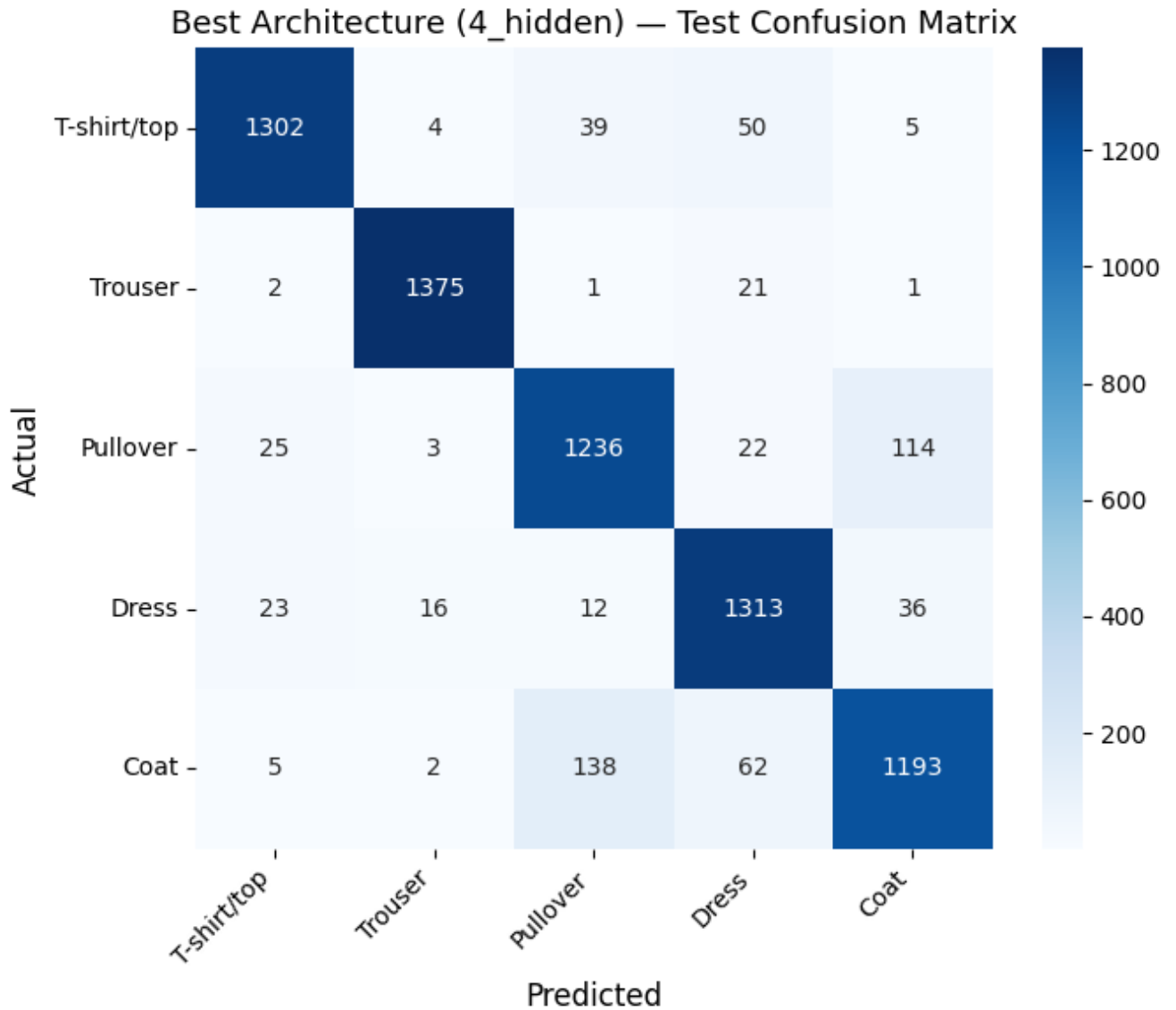


Figure 4: Test Confusion Matrix

The confusion matrices provide class-wise information about the predictions and show which clothing categories are most frequently confused with one another.

5.7 Architecture Comparison

For SGD, the validation accuracies of the three architectures were:

Table 8: Comparison of FCNN architectures using SGD

Architecture	Training Accuracy	Validation Accuracy
3-hidden	95.30%	90.80%
4-hidden	95.19%	91.41%
5-hidden	94.99%	91.20%

The four-hidden-layer architecture achieved the highest validation accuracy. Adding a fifth hidden layer did not further improve performance.

5.8 Optimizer Comparison

The best validation accuracies obtained by each optimizer across the three architectures were:

Table 9: Best validation accuracy for each optimizer

Optimizer	Best Validation Accuracy
SGD	91.41%
NAG	91.09%
Momentum	90.57%
Adam	90.41%
RMSProp	87.57%
Batch GD	20.66%

Among the tested configurations, SGD produced the highest validation accuracy.

6 Discussion

The experimental results demonstrate that both the network architecture and the optimization algorithm have an important influence on classification performance.

The four-hidden-layer FCNN trained using SGD was the best-performing model, obtaining a validation accuracy of 91.41% and a test accuracy of 91.70%. This indicates that the model learned useful representations from the pixel inputs and generalized effectively to previously unseen samples.

Increasing the depth from three hidden layers to four hidden layers improved the validation accuracy for SGD from 90.80% to 91.41%. However, increasing the depth further to five hidden layers resulted in a slightly lower validation accuracy of 91.20%. Therefore, increasing the number of layers did not automatically improve performance.

The training accuracy of the best model was 95.19%, while its validation accuracy was 91.41%. The difference is

$$95.19 - 91.41 = 3.78\%.$$

This difference indicates some degree of overfitting, although the test accuracy of 91.70% shows that the model still generalized well.

The optimizer comparison also produced several important observations. Although adaptive optimizers such as Adam and RMSProp can provide rapid parameter updates, SGD achieved the highest validation accuracy under the specific experimental conditions used here.

Batch Gradient Descent produced approximately 20% validation accuracy. Because the problem contains five balanced classes, 20% corresponds to the accuracy expected from random guessing. Thus, the model did not learn a useful classification function in the Batch GD experiments.

An important observation is that the convergence criterion does not necessarily indicate good classification performance. Batch GD was considered converged after six epochs because the change in average training error became smaller than the specified tolerance. Nevertheless, its classification accuracy remained close to chance level. This demonstrates that a small change in loss between epochs does not necessarily imply that the model has learned a good solution.

RMSProp also showed poor performance for some architectures. In particular, the 3-hidden-layer RMSProp model reached the maximum of 100 epochs without satisfying the convergence criterion and achieved only 55.41% validation accuracy.

These observations emphasize that optimizer performance depends on the architecture, learning rate, batch size, and other training conditions. Therefore, there is no single optimizer that is guaranteed to outperform all others in every neural-network problem.

7 Conclusion

A Fully Connected Neural Network was successfully implemented for five-class Fashion-MNIST classification. Three network architectures were evaluated using six optimization algorithms, resulting in a total of 18 experiments.

The main findings of the experiment are summarized below:

- A balanced dataset of 35,000 Fashion-MNIST images was created.
- The dataset was divided into 64% training, 16% validation, and 20% test sets.
- Three FCNN architectures containing three, four, and five hidden layers were evaluated.
- Six optimization algorithms were compared.
- The four-hidden-layer network trained using SGD achieved the highest validation accuracy of 91.41%.
- The selected model achieved a final test accuracy of 91.70%.
- Increasing the network depth beyond four hidden layers did not improve validation performance.
- Batch Gradient Descent achieved only approximately 20% accuracy, despite satisfying the numerical convergence criterion.
- RMSProp showed weak performance under the selected experimental settings for some architectures.

Overall, the experiment demonstrates that a relatively simple fully connected neural network can achieve strong performance on the selected Fashion-MNIST classification task. Under the experimental settings used in this study, the four-hidden-layer FCNN optimized using SGD provided the best combination of validation and test performance.